

## BIO DATA



Name : Dr.S. Kannan, BOYSCAST Fellow  
Designation : Professor & Head  
Department : EEE  
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### 1. Educational Qualification

| Degree | Branch / Specialization                                                                                                                                                                                                                                                            | Institute/University                   | Year of Passing |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|-----------------|
| B.E.   | Electrical and Electronics Engineering                                                                                                                                                                                                                                             | Madurai Kamaraj University             | 1991            |
| M.E    | Power System Engineering                                                                                                                                                                                                                                                           | Madurai Kamaraj University             | 1998            |
| Ph.D   | Power System Engineering                                                                                                                                                                                                                                                           | Madurai Kamaraj University             | 2005            |
| PDF    | Better Opportunities for Young Scientists In Chosen Areas of Science and Technology (BOYSCAST) fellowship – Awarded by Department of Science and technology (DST), Govt. of India, New Delhi<br><br>Multicriteria decision making including Genetic Algorithms and Neural networks | Iowa State University, Ames, Iowa, USA | 2007            |

### 2. Professional/ Industry Experience

: (in chronological order)

| S.No | Designation                                                             | Institution/ Organization                      | Period     |          |
|------|-------------------------------------------------------------------------|------------------------------------------------|------------|----------|
|      |                                                                         |                                                | From       | To       |
| 1.   | Professor, Head<br>Department of Electrical and Electronics Engineering | Ramco Institute of Technology,<br>Rajapalayam. | 09.06.2014 | till now |

Updated as on 02.04.2026

|     |                                                                           |                                                    |            |            |
|-----|---------------------------------------------------------------------------|----------------------------------------------------|------------|------------|
| 2.  | Sr.Professor, Department of Electrical and Electronics Engineering        | Kalasalingam University, Krishnankoil.             | 01.09 2012 | 07.06.2014 |
| 3.  | Dean (Admissions & Student Affairs)                                       | Kalasalingam University, Krishnankoil              | 01.07.2012 | 07.10.2013 |
| 4.  | Professor, Department of Electrical and Electronics Engineering           | Kalasalingam University, Krishnankoil.             | 01.01.2008 | 31.08.2012 |
| 5.  | Head of the Dep.t, Electronics and Communication Engineering              | Kalasalingam University, Krishnankoil.             | 15.11.2007 | 13.09.2009 |
| 6.  | Associate Professor                                                       | Kalasalingam University, Krishnankoil.             | 01.01.2006 | 31.12.2007 |
| 7.  | Head of the Dept., Instrumentation and Control Engineering                | Kalasalingam University, Krishnankoil.             | 23.09.2004 | 30.09.2006 |
| 8.  | Assistant Professor, Department of Electrical and Electronics Engineering | Kalasalingam University, Krishnankoil.             | 01.07.2005 | 31.12.2005 |
| 9.  | Senior Lecturer, Department of Electrical and Electronics Engineering     | Kalasalingam University, Krishnankoil.             | 01.01.2003 | 30.06.2005 |
| 10. | Lecturer, Department of Electrical and Electronics Engineering            | Kalasalingam University, Krishnankoil.             | 06.07.2000 | 31.12.2002 |
| 11. | Lecturer, Department of Electronics and Instrumentation Engineering       | Kamaraj College of Engg., and Tech., Virudhunagar, | 01.06.1999 | 5.07.2000  |

|     |                                                                           |                                                                  |            |            |
|-----|---------------------------------------------------------------------------|------------------------------------------------------------------|------------|------------|
| 12. | Lecturer, Department of Electrical and Electronics Engineering            | P.S.N.A.College of Engg., &Tech., Dindigul,                      | 22.12.1998 | 31.05.1999 |
| 13. | Part-time Lecturer, Department of Electrical and Electronics Engineering, | Thiagarajar College of Engg., Madurai,                           | Dec.1997   | Nov.1998   |
| 14. | Lecturer, Department of Electrical and Electronics Engineering            | Lakshmi Ammal Polytechnic, Kovilpatti, Tamilnadu, India          | 15.07.1996 | 05.08.1997 |
| 15. | Maintenance Engineer                                                      | E.C.E Industries Ltd., Elevator division, Cochin, Kerala, India. | 27.01.1994 | 13.07.1996 |
| 16. | Graduate Engineer Trainee                                                 | Tamilnadu Electricity, Board, Virudhunagar, Tamilnadu, India     | 22.07.1992 | 21.07.1993 |

### 3. Research Interest

Power System Engineering

### 4 Publications

#### 4.1 JOURNAL PUBLICATIONS

##### 2003

1) S. Kannan, S. Jawahar, S. Mary Raja Slochanal and Narayana Prasad Padhy, “Generation expansion planning with power flow constraints-Evolutionary strategy approach”, Water and Energy International Journal, Central Board of Irrigation and power, NewDelhi, Vol. 60, No. 3, pp. 37-44, July-September 2003, Print ISSN: 0974-4207, Online ISSN:0974-4711. <http://www.indianjournals.com/ijor.aspx?target=ijor:wei&volume=60&issue=3&article=005>

##### 2004

2) S. Kannan, S. Mary Raja Slochanal, S. Jawahar and Narayana Prasad Padhy, “Genetic algorithm approach to generation expansion planning under deregulated environment”, Journal of Energy and Environment, Centre for Energy Studies, BUET, Dhaka-1000, Bangladesh, pp. 1-13, Vol. 3, May 2004, ISSN: 1563-1362.

3) S. Kannan, S. Mary Raja Slochanal, P. Subbaraj and Narayana Prasad Padhy, “Application of Particle swarm optimization technique and its variants to Generation expansion planning problem”, Electric Power Systems Research, (Elsevier Science Ltd) Vol. 70, Issue 3, pp. 203-210, August 2004. *IF:1.259*, ISSN: 0378-7796

##### 2005

4) S. Kannan, S. Mary Raja Slochanal and Narayana Prasad Padhy, “Application and comparison of metaheuristic techniques to generation expansion planning problem”, IEEE Transactions on Power Systems, Vol. 20, No. 1, pp 466-475, Feb 2005, *IF:1.938*, ISSN: 0885-8950.

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5) K. Mahadevan, P.S. Kannan, S. Kannan, “Particle Swarm Optimization for Economic Dispatch of generating units with valve-point loading”, Journal of Energy and Environment, Centre for Energy Studies, BUET, Dhaka – 1000, Bangladesh, Vol. 4, pp. 49-61, May 2005, ISSN: 1563-1362.

6) S. Kannan, S. Mary Raja Slochanal, Narayana Prasad Padhy and S. Suba, “Improved Evolutionary Programming Based Generation Expansion Planning under Partially Deregulated Environment”, Journal of International Association on Electric Generation Transmission and Distribution (Afro-Asian Region), Central Board of Irrigation and Power (CBIP), New Delhi, Vol. 17-18 No. 1, March - June 2005, Online ISSN: 2229-4449.

## **2006**

7) S. Kannan, S. Mary Raja Slochanal, S. Baskar and Narayana Prasad Padhy, “Application and comparison of metaheuristic techniques to generation expansion planning incorporating both bilateral and multilateral transactions”, International Journal of Emerging Electric Power Systems, Vol. 6, Issue 1, Article 2, 2006, pp.1-12, *IF:0.224*, ISSN: 1553 - 779 X

## **2007**

8) S. Kannan, S. Mary Raja Slochanal, S. Baskar and P.Murugan, “Application and comparison of metaheuristic techniques to generation expansion planning in the partially deregulated environment”, IET Generation, Transmission & Distribution, Volume 1, Issue 1 , January 2007, pp. 111-118. *IF:0.856*, ISSN. Print: 1751-8687; Online: 1751-8695.

## **2009**

9) S.Kannan, and P.Murugan, “Solutions to transmission constrained generation expansion planning using Differential evolution,” European transaction on electrical power, Wiley Interscience, Vol. 19, pp1033-1039, 2009. *IF: 0.32*, ISSN:1546-3109.

10) S.Kannan, S. Baskar, James D McCalley and P. Murugan, “Application of NSGA-II Algorithm to generation expansion planning,” IEEE Trans. on power system, Vol. 24, No. 1, pp 454-461, Feb. 2009. *IF:1.938*, ISSN: 0885-8950.

11) P. Murugan, S.Kannan and S. Baskar, “NSGA-II Algorithm to Multi-objective Generation Expansion Planning Problem”, Electric Power Systems Research, Vol.79, No.4, pp 622-628, April 2009. *IF:1.259*, ISSN: 0378-7796

12) S. Ramesh, S. Kannan, and S. Baskar, “Application of a Fast and Elitist Multi-Objective Genetic Algorithm to Reactive Power Dispatch”, Serbian Journal of Electrical Engineering, Vol.6, No.1, May 2009, pp.119-133. ISSN 1451 – 4869

13) P. Murugan, S. Kannan and S. Baskar, “Application of NSGA-II Algorithm to Single-Objective Transmission constrained Generation Expansion Planning”, IEEE Trans. on power system, Vol. No. 24, No.4, pp 1790-1797, Nov. 2009. *IF:1.938*, ISSN: 0885-8950.

## **2011**

14) S. Dhanalakshmi, S. Kannan, K. Mahadevan, S. Baskar, “Application of Modified NSGA-II Algorithm to Combined Economic and Emission dispatch problem”, International Journal of Electrical Power and Energy Systems, Vol. 33, 2011, pp. 992-1002. *IF:1.61*, ISSN: 0142-0615

## **2012**

15) S.Ramesh, S.Kannan and S.Baskar, “Application of Modified NSGA-II Algorithm to Multi-objective Reactive Power Planning”, Applied Soft Computing – Elsevier, Vol. 12, Issue 2, Feb. 2012, pp. 741-753, *IF: 2.084*

16) S. Ramesh, S. Kannan, and S. Baskar, "An Improved Generalized Differential Evolution Algorithm for Multi-objective Reactive Power Dispatch", Taylor and Francis- Engineering Optimization, Vol.44, No.4, April 2012, pp.391-405, *IF: 0.902* (Print ISSN: 0305-215x, Online ISSN: 1029-0273).

17) B. Rampriya, K. Mahadevan, and S. Kannan, "A Hybrid LR-FA Technique to Optimize the Profit Function of GENCOS in a Restructured Power System", International Journal of Power and Energy Systems, Vol. 32, No. 3, pp. 113-120, 2012 (Acta Press), ISSN (Online): 1710-2243, ISSN (Hardcopy): 1078-3466

18) A.Muthukumar, C.Kasthuri, S.Kannan, "Multimodal Biometric Authentication Using Particle Swarm Optimization Algorithm with Fingerprint and Iris" *ICTACT Journal on Image and Video Processing (IJIVP)*, An international publication of ICT Academy of Tamilnadu, Vol. 02, Issue 03, Feb. 2012, pp.369-374 (ISSN 0976-9099).

19) B.Rampriya, K.Mahadevan and S.Kannan, "Application of Differential Evolution to Dynamic Economic Dispatch problem with transmission losses under various bidding strategies in electricity markets", Journal of Electrical Engineering & Technology, JEET, Vol.7, No.5, September 2012, pp.681-688, <http://dx.doi.org/10.5370/JEET.2012.7.5.681>; IF 0.579

20) S. Saravanan, S. Kannan, C. Thangaraj, "India's Electricity Demand forecast using Regression Analysis and Artificial Neural Networks based on principal components", ICTACT Journal on Soft Computing (IJSC), An international publication of ICT Academy of Tamilnadu, July 2012, Vol. 2, Issue 4, pp.365-370

21) K. Karthikeyan, S. Kannan, S. Baskar and C. Thangaraj, "Application of Restart Covariance Matrix Adaptation Evolution Strategy (RCMA-ES) to Generation Expansion Planning Problem", "ICTACT Journal on Soft Computing" Volume -3, Issue-1, October 2012, pp.401-407

22) S. Arun Singh, A.Muthukumar, S.Kannan, "Iris Authentication using SIFT with SVM" *Special Issue of International Journal of Computer Applications (0975 – 8887) on International Conference on Electronics, Communication and Information Systems (ICECI 12)*, Vol. 02, November 2012, pp.1-3 (ISBN 973-93-80870-65-2).

23) B. Sivasubramanian, K. Vijayakumar, and S. Kannan, "Design And Implementation of Passive Filter In Industrial Applications" i-manager's Journal on Instrumentation & Control Engineering, Vol. 1, No. 1, November 2012 - January 2013, pp.18-24

## **2013**

24) S.Dhanalakshmi, S.Baskar, S.Kannan and K.Mahadevan, "Generation Scheduling Problem by Intelligent Genetic Algorithm", Computer and Electrical Engineering, (Elsevier), Vol. 39, Issue. 1., January 2013, pp. 79-88.( *IF: 0.5*), ISSN: 0045-7906.

25) K. Karthikeyan, S. Kannan, S. Baskar and C. Thangaraj, "Application of Opposition based DE to Generation Expansion Planning Problem", Journal of Electrical Engineering and Technology, Vol. 8, No. 4, pp. 686-693, 2013, ISSN(Print) 1975-0102, ISSN(Online) 2093-7423, <http://dx.doi.org/10.5370/JEET.2013.8.4.686>

26) A.Muthukumar and S.Kannan, "Finger Knuckle print Recognition with SIFT and K-Means Algorithm", *ICTACT Journal on Image and Video Processing (IJIVP)*, An International publication of ICT Academy of Tamilnadu, Vol. 03, Issue 03, ISSN 0976-9099, February 2013, pp.583-588

27) M.Rajkumar, K.Mahadevan, S.Kannan, and S.Baskar, "Combined economic and emission dispatch with valve-point loading of thermal generators using modified NSGA-II", Journal of Electrical Engineering & Technology (JEET), Vol.8, No.3, May 2013, pp. 490-498. <http://dx.doi.org/10.5370/JEET.2013.8.3.490>, ISSN (Print) 1975-0102, ISSN(Online) 2093-7423.

28) K. Karthikeyan, S. Kannan, S. Baskar and C. Thangaraj, "Application of Self-adaptive Differential Evolution Algorithm to Generation Expansion Planning Problem", Journal of Electrical Systems, Volume 2, Issue 2, 2013, pp. 203-211.

29) R.Krishna Priya, C.Thangaraj, C. Kesavadas and S. Kannan, "Fuzzy Entropy based MR Brain Image Segmentation Using Modified Particle Swarm Optimization" International Journal of Imaging Systems and Technology (Wiley Blackwell), Vol. 23, Issue 4, Dec. 2013, pp. 281- 288, (Impact Factor: 0.799).

## **2014**

30) S.Ponnayira Sundaravel, K.Vijayakumar and S.Kannan, "Reactive Power Control Using STATCOM", "IEEE International Conference on Emerging Trends in Science, Engineering, Business and Disaster Management" [ICBDM 2014] on 28th February 2014, Journal of Emerging Technologies, Image Processing & Networking, Vol.8, Special Issue:1, Feb 2014, ISSN No.: 0973-2993, Published by The Department of Electrical and Electronics Engineering, Noorul Islam Centre for Higher Education, Noorul Islam University, Nagercoil, INDIA, pp. 288-293.

31) T.Arunprasath, M.Pallikonda Rajasekaran and S.Kannan, "A Quantitative Assessment of PET Brain Image Reconstruction using MAP and Neural Network based Segmentation of CG Algorithm", International Journal of Computer Information Systems and Industrial Management Applications, Vol. 6, 2014, pp. 381- 390. ISSN 2150-7988, [www.mirlabs.net/ijcisim/index.html](http://www.mirlabs.net/ijcisim/index.html)

32) M.Rajkumar, K.Mahadevan, S.Kannan, and S.Baskar, "NSGA-II technique for Multiobjective generation dispatch of thermal generators with non-smooth fuel cost functions" Journal of Electrical Engineering & Technology (JEET), Vol. 9, No. 2: 423-432, 2014, <http://dx.doi.org/10.5370/JEET.2014.9.2.742>, ISSN(Print) 1975-0102 ISSN(Online) 2093-7423.

33) A.Muthukumar and S.Kannan, "K-Means Based Multimodal Biometric Authentication using Fingerprint and Finger Knuckle Print with Feature level fusion", Iranian Journal of Science & Technology, IJST, Transactions of Electrical Engineering, Vol. 37, No. E2, pp. 133-145 (SCI Impact Factor: 0.6), 2013

34) K.Karunanithi, S.Kannan, C.Thangaraj, "Generation Expansion Planning for Tamil Nadu-A case study", International Transactions on Electrical Energy Systems, vol. 25, Issue 9, pp. 1771-1787, DOI: 10.1002/etep.1929, March 2014. ISSN: 2050-7038.

35) J.Kohila, R.Muniaraj, and S.Kannan, "Z-Source Multilevel Inverter for Photovoltaic Application", Special Issue of International Journal of Innovative Research in Science, Engineering and Technology (IJIRSET), Vol.03, Special Issue 03, March 2014, pp.492-497, ISSN (Print): 2347-6710, ISSN (Online): 2319-8753, M.R.Thanasekar and N.Balaji (Eds.): ICIET 2014

(IEEE International conference on Innovation in Engineering and Technology, ICIET-2014, held on 21-22 March 2014, K.L.N college of engineering, Madurai, Tamilnadu, pp.473-478)

36) Arunprasath, T., Pallikonda, R. M. and Kannan, S. (2014), “PET image reconstruction using ANN”, *International Journal of Imaging Systems and Technology*, vol. 24, Issue 3, pp. 249–255, Sep. 2014, doi: 10.1002/ima.22100, (Impact Factor: 0.768), ISSN: 1098-1098

37) S.Suresh, S.Kannan and S.Ponnayira Sundaravel, “A Modified Single phase H-Bridge Inverter for High Power Applications with Power Quality Improvement”, *Applied Mechanics and Materials* Vol. 626 (2014) pp.118-126, © (2014) Trans Tech Publications, Switzerland, doi:10.4028/www.scientific.net/AMM.626.118

38) S.Saravanan, S.Kannan and C.Thangaraj, “Electric energy consumption of domestic sector in INDIA”, *REVISTA TECNICA DE LA FACULTAD DE INGENIERIA UNIVERSIDAD DEL ZULIA* (Technical Journal of the Faculty of Engineering, TJFE), Rev. Téc. Ing. Univ. Zulia. Vol. 37, No. 3, pp.41-50, 2014. (Impact factor - 0.033)

39) J.Booma, K.Mahadevan and S.Kannan, “Selection of optimal capacity of wind turbine generator for Generation Expansion Planning by considering cost and reliability”, *International Journal of Applied Engineering Research (IJAER)*, *Special Issue*, Vol. 9, No. 24, 2014, pp. 8403-8406

40) S.Saravanan, S.Kannan, C.Thangaraj, “Predicting India’s electricity consumption: A comparison of least square support vector machine and artificial neural networks”, *International Journal of Applied Engineering Research (IJAER)*, *Special Issues*, Vol. 9, No. 26, 2014, pp. 9094-9097

41) T.Arunprasath, M.Pallikonda Rajasekaran, S.Kannan, E.Malarvizhi, “A Quantitative Assessment of PET Image Reconstruction using Filtered Back Projection (FBP), Conjugate Gradient (CG) and Artificial Neural Network (ANN) Algorithm”, *International Journal of Applied Engineering Research (IJAER)*, *Special Issues*, Volume 9, Number 26, 2014, pp. 9107-9110

## 2015

42) M.Rajkumar, K.Mahadevan, S.Kannan and S.Baskar, “Environmental and economic power dispatch of thermal generators using modified NSGA-II algorithm”, *International Transactions on Electrical Energy Systems (Formerly European Transactions on Electrical Power)*, Vol. 25, Issue 8, 2015, pp. 1552-1569. - DOI: 10.1002/etep.1918.

2015\_MR\_KM\_SK\_SB\_Env and eco power dispatch of thermal gen. using modified NSGA-II algorithm”, *International Transactions on Electrical Energy Systems (Formerly European Transactions on Electrical Power)*, Vol. 25, Issue 8, 2015, pp. 1552-1569. - DOI: 10.1002/etep.1918.

43) A.Muthukumar, S.Kannan “Finger Knuckle Print Authentication using AES and K- Means Algorithm”, *International Arab Journal of Information and technology*, vol. 12, no. 6A, December 2015, pp. 642-649

44) A.Muthukumar, S.Kannan, “AES Based Multimodal Biometric Authentication using Cryptographic Level Fusion with Fingerprint and Finger Knuckle Print”, *International Arab Journal of Information and technology*, vol. 12, No. 5, Sep. 2015, pp.431-440.

45) T.Arunprasath, M.Pallikonda Rajasekaran and S.Kannan, “ANFIS - EM approach For PET brain image reconstruction”, *Int. J. Imaging Syst. Technol.*, vol.25, no.1, pp. 1-6. March 2015, doi: 10.1002/ima.22114

- 46) T.Arunprasath, M.Pallikonda Rajasekaran and S.Kannan, "Neural network segmented CD algorithm-based PET liver image reconstruction", *Int. J. Biomedical Engineering and Technology*, vol. 17, no. 3, January 2015 pp.276-289, Inderscience publications.
- 47) S.Saravanan, G.Saravanakumar, S.Kannan, C.Thangaraj, "Electric energy consumption of industrial sector in India", *International Journal of Applied Engineering Research (IJAER), Special Issues*, 2015, vol. 10, no. 6, 2015, pp. 5044-5049.
- 48) S.Saravanan, S.Amosedinakaran, S.Kannan, C.Thangaraj, "Estimating electric energy consumption: Case study of India", *International Journal of Applied Engineering Research (IJAER), Special Issues*, vol. 10, no. 6, 2015, pp. 5298-5304.
- 49) J. Kohila, V. Suresh Kumar, and S.Kannan, "ICOS $\phi$  algorithm based control for distribution static compensator", *International Journal of Applied Engineering Research (IJAER), Special Issues*, vol. 10, no. 6, 2015, pp. 5553- 5558
- 50) B.R.Prabakar, K.Karunanithi, S.Kannan, C. Thangaraj, "Energy model of electric sector for Tamil Nadu", *International Journal of Applied Engineering Research (IJAER), Special Issues*, vol. 10, no. 6, 2015, pp. 5681- 5687
- 51) N.Pothirasan, K.Vijayakumar, S.Ponnayirasundaravel, S.Kannan, "Reactive Power Compensation Using Cascaded Multi Level Inverter", *International Journal of Applied Engineering Research (IJAER), Special Issues*, vol. 10, no. 9, 2015, pp. 7452-7456.
- 52) R. Krishna Priya, C.Thangaraj, C.Kesavdas, S.Kannan, "Brain Tumor Segmentation in MRI Images Using Integrated Modified PSO-Fuzzy Approach", *International Arab Journal of Information Technology*, 12 (6A), 2015, 797-804.
- 53) D.Danalakshmi and S.Kannan, "Market based reactive power cost allocation in restructured power system", *REVISTA TECNICA DE LA FACULTAD DE INGENIERIA UNIVERSIDAD DEL ZULIA (Technical Journal of the Faculty of Engineering, TJFE), Rev. Téc. Ing. Univ. Zulia. Vol. 38, No. 1, pp.97-104, April 2015. (Impact factor - 0.033)*
- 54) S. Saravanan, S. Kannan, and C. Thangaraj, "Prediction of India's electricity demand using ANFIS", *ICTACT Journal on Soft Computing (IJSC)*, An international publication of ICT Academy of Tamil Nadu, vol. 05, Issue. 03, April 2015, pp. 985-990.
- 55) J. Booma, K. Mahadevan and S. Kannan, "Minimum cost estimation of generation expansion planning incorporating wind power plant", *ARPN Journal of Engineering and Applied Sciences*, Asian Research Publishing Network (ARPN), vol. 10, no.7, April 2015, pp.2956-2961.
- 56) K. Karthikeyan, S. Kannan, S. Baskar and C. Thangaraj, "Application of NSGA-II, Modified NSGA-II and IGDE-3 Algorithms to Multi-Objective Transmission Constrained Generation Expansion Planning", *International Journal of Applied Engineering Research (IJAER), Special Issues*, vol. 10, no. 44, 2015, pp. 31004 – 31013.
- 57) K.Karthikeyan, S.Kannan, S.Baskar, C.Thangaraj, "Application of MOEA/D Algorithm to Multi-Objective Generation Expansion Planning", *International Journal of Applied Engineering Research (IJAER), Special Issues*, vol. 10, no. 44, 2015, pp. pp.31875-31882.
- 58) P. Iniyatharasi, M. Pallikonda Rajasekaran, T. Arun Prasath and S. Kannan, "PET Image Reconstruction Using ISRA Technique", *Australian Journal of Basic and Applied Sciences*, vol. 9, no.16, 2015, pp. 110-117.



59) K.Karunanithi, S.Kannan and C.Thangaraj, “Reliability analysis of Tamil Nadu power grid for the year 2013”, International Journal of Applied Engineering Research (IJAER), *Special Issues*, vol. 10, no. 55, 2015, pp. 3032-3038.

60) D.Danalakshmi and S.Kannan, “Market based pricing for reactive power service in restructured practical power system”, International Journal of Applied Engineering Research (IJAER), *Special Issues*, vol. 10, no. 55, 2015, pp. 3137-3142.

61) P.Shanmugapriya, T.Arunprasath and S.Kannan, “PET Image Reconstruction Using Extreme Learning Machine”, International Journal of Applied Engineering Research (IJAER), *Special Issues*, vol. 10, no. 59, 2015, pp. 387-389.

62) S.Suresh, S.Kannan and P.Murugan, “Power Quality Improvement using Asymmetrical Multilevel Inverter for Grid Connected Photo Voltaic System”, REVISTA TECNICA DE LA FACULTAD DE INGENIERIA UNIVERSIDAD DEL ZULIA (Technical Journal of the Faculty of Engineering, TJFE), Rev. Téc. Ing. Univ. Zulia. Vol. 38, No. 3, pp.47-63, December 2015. (Impact factor - 0.033)

63) K.Karunanithi, A.Bhuvanesh, S.Kannan, “Generation Expansion Planning - A Tutorial Paper”, VFSTR Journal of STEM, vol.01, no.02 (2015) 2455-2062, pp. 1-11

64) M. Rajkumar, K. Mahadevan and S. Kannan, “Evolutionary programming solution for Economic Dispatch of units with ramp rate limits”, VFSTR Journal of STEM, vol.01, no.02 (2015) 2455-2062, pp. 12-17

65) K.Rajesh, K.Karthikeyan, S.Kannan and M.Karuppasamypandian, “Generation Capacity Expansion Planning with Solar Power Plant Incorporating Emission”, VFSTR Journal of STEM, vol.01, no.02 (2015) 2455-2062, pp.18-37

## 2016

66) K.Rajesh, A.Bhuvanesh, S.Kannan, C.Thangaraj “Least Cost Generation Expansion Panning with Solar Power Plant using Differential Evolution Algorithm” “Renewable Energy” (Elsevier), vol. 85, January 2016, pp.677-686. DOI: <http://dx.doi.org/10.1016/j.renene.2015.07.026>. (SCI)

67) K.Rajesh, K.Karthikeyan, S.Kannan and C.Thangaraj, “Generation Expansion Planning based on Solar plants with Storage”, Renewable and Sustainable Energy Reviews (Elsevier), vol. 57, 2016, pp.953-964, DOI: 10.1016/j.rser.2015.12.126 (SCOPUS) (SCI)

68) K.Rajesh, S.Kannan and C.Thangaraj, “Least Cost Generation Expansion Planning With Wind Power Plant Incorporating Emission Using Differential Evolution Algorithm”, International Journal of Electrical Power and Energy Systems (Elsevier), vol. 80, 2016, pp.275-286, doi:10.1016/j.ijepes.2016.01.047 (SCOPUS) (SCI)

69) M. ArumugaBabu, R.Mahalakshmi, S.Kannan, M.Karuppasamypandian and A.Bhuvanesh, “Application of Self Adaptive Differential Evolution Algorithm for Optimal Placement and Sizing of Renewable DG Sources in Distribution Network Including Different Load Models”, International journal of advanced engineering technology, Int J AdvEngg Tech/Vol. VII/Issue I, pp. 668-675, Jan.-March, 2016.

70) M.Arumuga Babu, R.Mahalakshmi, S.Kannan, M.Karuppasamypandian, A.Bhuvanesh, “Economic Dispatch of Distributed Generation Using Backtracking Search Optimization Algorithm”,

71) J.Booma, K.Karunanithi, K.Mahadevan and S.Kannan, "Reliability analysis for restricted and unrestricted demand of Tamilnadu power grid for the year 2014 using WASP-IV", International Journal of Future Innovative Science and Engineering Research (IJFISER), vol.02, Issue-II, June 2016, ISSN (Online): 2454- 1966, [www.istpublications.com](http://www.istpublications.com).

72) S. Suresh, S. Kannan, R. Karthigaivel, "Reactive Power Compensation using Fuzzy Logic Controlled Multilevel Inverter for Power Distribution System", Journal of Computer Science and Control Systems (JCSCS), vol. 9, no. 2 October 2016, pp.41-47.

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## 4.3 Book/Book chapter publications

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3. K.Vijayakumar, R. Sundar raj, and S. Kannan, “Application of Sinusoidal Pulse Width Modulation Based Matrix Converter as revolutionized Power Electronic Converter”, Lecture notes in electrical engineering, *Power Electronics and Renewable Energy Systems*. Springer India, 2015, Series Volume 326, Chapter 2, pp.9-17, DOI 10.1007/978-81-322-2119-7\_2, (The International Conference on Power Electronics and Renewable Energy Systems, Proceedings of ICPERES 2014, held at Rajalakshmi



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  5. K.Karunanithi, S.Kannan, C.Thangaraj, "Reliability Evaluation of Tamil Nadu Power Grid for the year 2012", Lecture notes in electrical engineering, *Power Electronics and Renewable Energy Systems*. Springer India, 2015, Series Volume 326, Chapter 104, pp.1067-1076, DOI 10.1007/978-81-322-2119-7\_104, (The International Conference on Power Electronics and Renewable Energy Systems, Proceedings of ICPERES 2014, held at Rajalakshmi Engineering College, Chennai, India, in association with Springer and IEEE, 24-26 April, 2014).
  6. S.Saravanan, S.Kannan, C.Thangaraj, "Prediction of India's Industrial Sector Electricity consumption using ANFIS", Lecture notes in electrical engineering, *Power Electronics and Renewable Energy Systems*. Springer India, 2015, Series Volume 326, Chapter 124, pp.1281-1288, DOI 10.1007/978-81-322-2119-7\_124. (The International Conference on Artificial Intelligence and Evolutionary Algorithms in Engineering Systems, ICAEES 2014, Noorul Islam Centre for Higher Education, Kumaracoil, Thuckalay, Kanyakumari district, Tamilnadu - 629180, INDIA, held on 23-24 April 2014).
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  8. S.Dhanalakshmi, S.Kannan, S.Baskar and K.Mahadevan, "Multi-objective Generation Scheduling using Modified Non-dominated Sorting Genetic Algorithm-II", In *Swarm, Evolutionary, and Memetic Computing*, pp. 456-470. Springer International Publishing, 2015, B.K. Panigrahi et al. (Eds.): SEMCCO 2014, LNCS 8947, pp. 456–470, 2015, DOI: 10.1007/978-3-319-20294-5\_40, 5<sup>th</sup> Joint International Conference on Swarm, Evolutionary and Memetic Computing (SEMCCO 2014), 18-20 December 2014, organized by Siksha 'O' Anusandhan University, Bhubaneswar, Odisha, India.
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  10. S. Ramesh, S. Kannan and S.Baskar, "Application of Modified NSGA-II Algorithm to Reactive Power Optimization, **Book Title:** Trends in Computer Science, Engineering and Information Technology, **Book Subtitle:** First International Conference on

- Computer Science, Engineering and Information Technology, CCSEIT 2011, Tirunelveli, Tamil Nadu, India, September 23-25, 2011. Proceedings, Nagamalai, D., Renault, E. and Dhanushkodi M. Eds. CCIS 204, Springer-Verlag, Berlin Heidelberg, Volume 204, 2011, pp. 344–354, Sep 23-25, 2011, Organised by Manonmaniam Sundharanar University, Tirunelveli-627012, Tamilnadu, India. **DOI:** 10.1007/978-3-642-24043-0\_35 **Print ISBN:** 978-3-642-24042-3, **Online ISBN:** 978-3-642-24043-0
11. S. Dhanalakshmi, S. Kannan, S. Baskar and K. Mahadevan, “Intelligent Genetic Algorithm for Generation Scheduling under deregulated environment”, SEMCCO 2011, 19-21, Dec. 2011, Anil Neerukonda Institute of Technology & Sciences, Visakhapatnam, Andhra Pradesh, India Panigrahi B.K. et al. Eds. SEMCCO 2011, Part I, Lecture Notes in Computer Science (LNCS) 7076, Springer-Verlag, Berlin Heidelberg, pp. 282-289, 2011. (SNIP: 0.020 and SJR: 0.084).
  12. S.Ramesh, S.Kannan and S.Baskar, “Application of an Improved Generalized Differential Evolution algorithm to multi-objective optimization problems”. SEMCCO 2011, 19-21, Dec. 2011, Anil Neerukonda Institute of Technology & Sciences, Visakhapatnam, Andhra Pradesh, India. Panigrahi B.K. et al. Eds. SEMCCO 2011, Part I, Lecture Notes in Computer Science (LNCS) 7076, Springer- Verlag, Berlin Heidelberg, pp. 77–84, 2011. (SNIP: 0.020 and SJR: 0.084).
  13. Short notes were prepared for EE8702 Power System Operation and Control & EE3601 Protection and Switchgear course.
  14. Dr. S. KANNAN, "Neural Network and Fuzzy Systems: Design Fundamentals", N/A, December 2024, International Publisher
  15. Dr. J Gnanavadeivel Dr. S. Kannan Dr. A Arunkumar, Dr. S. Jeyanthi, , “ Basic Electrical and Electronics Engineering , 205, Anuradha Publication( ISBN: 978-93-91677-82-4)

## 5.Funded Projects

| S.No. | Title of the project                                                                         | Funding Agency                 | Duration | Amount    | Status   |
|-------|----------------------------------------------------------------------------------------------|--------------------------------|----------|-----------|----------|
| 1.    | InstaClean AI VisionBot                                                                      | MSME 5.0                       | 1 year   | 15,00,000 | Applied  |
| 2.    | "SMART VISION-BASED DRIVER ASSISTANCE WITH ADVANCED NOTIFICATION SYSTEM"                     | AICTE RPS 2025 (Lab to Market) | 3 years  | 48,00,000 | Applied  |
| 3.    | customized tricycle assistive technology for people with disabilities                        | IE(I)                          | 9 Months | 30000/-   | Accepted |
| 4.    | Business-Plan-for-the-Cataract-Surgeons-Anterior-Vitrectomy-System-Using-Wireless-Connection | MSME                           | 1 year   | 15,00,000 | Granted  |
| 5.    | IoT-based Smart                                                                              | TNSCST                         | 2 years  | 5,00,000  | Applied  |

Updated as on 02.04.2026

|     |                                                                                                                                                      |                                                                                            |           |            |         |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------|------------|---------|
|     | System for Managing Food Waste                                                                                                                       |                                                                                            |           |            |         |
| 6.  | Risk Assessment of Rabies Transmission in Rural and Urban Areas                                                                                      | TNSCST                                                                                     | 3 years   | 5,00,000   | Applied |
| 7.  | Wireless Charging System for E-Vehicle                                                                                                               | IE(I)                                                                                      | 9 Months  | 32,000     | Applied |
| 8.  | Design and Development of an Integrated Ophthalmic Workstation Combining Surgical, Diagnostic, and Ergonomic Tools for Enhanced Eye Care Efficiency. | ICMR                                                                                       | 36 months | 21,714,240 | Applied |
| 9.  | New Education Policy 2020                                                                                                                            | INDIAN COUNCIL OF SOCIAL SCIENCE RESEARCH (ICSSR Short Empirical Research proposal.)       | 6 months  | 19,28,140  | Applied |
| 10. | Export of Power to Grid from Solar and Improving Power Quality using STATCOM                                                                         | DST-CSRI                                                                                   | 3 Yeras   | 1461400    | Applied |
| 11. | Wind Energy Forecasting for Tamil Nadu using Machine learning algorithms                                                                             | Ministry of New and Renewable Energy                                                       | 36 Months | 45,31,640  | Applied |
| 12. | IoT based food waste monitoring system                                                                                                               | Institution of Engineers-India(IE(I))                                                      | 9 months  | 20,000     | Granted |
| 13. | Design And Establishment Of Multi Source Renewable Energy Based E-Vehicle Charging Station And Management System                                     | TN Chief Minister's Research Grant (CMRG) Scheme by the Directorate of Technical Education | 3 years   | 39,77,000  | Applied |
| 14. | Agroforestry Digitalization With Data Science On                                                                                                     | TN Chief Minister's Research Grant (CMRG) Scheme                                           | 3 years   | 39,56,800  | Applied |

Updated as on 02.04.2026

|     |                                                                                                                   |                                                                                                                                                                                                             |           |                |         |
|-----|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------|---------|
|     | Climate Change                                                                                                    | by the Directorate of Technical Education                                                                                                                                                                   |           |                |         |
| 15. | Machine learning algorithms for wind energy forecasting in coastal regions of Tamilnadu                           | SERB – SUPRA                                                                                                                                                                                                | 3 years   | 51,41,304      | Applied |
| 16. | Portable Solar Drying System for Harvesting Sesame Seed in Alankulam Village                                      | "UNNAT BHARAT ABHIYAN For Technology Customization"                                                                                                                                                         | 6 Months  | 100000         | Applied |
| 17. | Design and Implementation of Battery swapping system for E vehicle                                                | DST SERB                                                                                                                                                                                                    | 36 months | Rs.25,79,000   | Applied |
| 18. | LoRA based waste water monitoring system in Textile Industries                                                    | Optimal Water Use in Industrial Sector-2021 - Research Stream - Establishment of Proof of Concept (POC) - Technology Mission Division (Energy, Water & Others) – Department of Science and Technology (DST) | 2 years   | Rs.17,58,900   | Applied |
| 19. | Grid connected Solar PV System with Battery – Supercapacitor Hybrid Energy Storage for rural households           | DST SERB                                                                                                                                                                                                    | 3 years   | 22,50,494      | Applied |
| 20. | A study on the impact of charging patterns of electric vehicles and solar PV on the Generation Expansion Planning | DST-Matrices                                                                                                                                                                                                | 36 months | Rs. 6,00,000/- | Applied |

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|-----|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|-------------------------|------------|-----------|
| 21. | Effi Cycle '20                                                                                                             | TRCL                                             | 1 year                  | 1.75       | Completed |
| 22. | Electric Solar Vehicle Championship '20                                                                                    | TRCL                                             | 1 Year                  | 2          | Completed |
| 23. | E-BAJA 2020                                                                                                                | TRCL                                             | 1 Year                  | 2          | Completed |
| 24. | Electric Two Wheeler Design Competition '20                                                                                | TRCL                                             | 1 Year                  | 0.4        | Completed |
| 25. | Electric Go-Kart Championships – 2020                                                                                      | TRCL                                             | 1 Year                  | 0.9        | Completed |
| 26. | SAEISS bicycle design competition 2020                                                                                     | TRCL                                             | 1 Year                  | 0.2        | Completed |
| 27. | M-BAJA 2020                                                                                                                | TRCL                                             | 1 Year                  | 2.25       | Completed |
| 28. | SAEISS Tractor Design Competition '20                                                                                      | TRCL                                             | 1 Year                  | 2.5        | Completed |
| 29. | “Development of Hybrid Metaheuristic Algorithms for solving Power System Operation Planning under deregulated Environment” | Department of Science and Technology, New Delhi, | Sep. 2007 to Sep. 2010. | 9,90,720/- | Completed |
| 30. | Autonomous Operated Robot for Water Tank Cleaning                                                                          | Institution of Engineers-India(IE(I))            | 0.5 year                | 0.45 lakhs | Completed |

#### 6.Ph.D. guidance

| Sl.No | Scholar name    | University                             | Topic                                                                                                                     | Status    |
|-------|-----------------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-----------|
| 1.    | S. Dhanalakshmi | Kalasalingam University, Krishnankoil. | “Application of Genetic Algorithms to Generation Scheduling under Regulated and Deregulated Environment”                  | Completed |
| 2.    | S. Ramesh,      | Anna University, Chennai.              | “Application and Comparison of Multi-Objective Evolutionary Algorithms to Reactive Power Planning and Dispatch Problems,” | Completed |

|     |                  |                                        |                                                                                                                                   |           |
|-----|------------------|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------|
| 3.  | A. Muthukumar    | Kalasalingam University, Krishnankoil. | “Secured integrated authentication system using multimodal biometrics with fusion techniques”                                     | Completed |
| 4.  | K. Karunanithi   | Kalasalingam University, Krishnankoil. | “Investigation of Power System Reliability and Generation expansion planning”                                                     | Completed |
| 5.  | S. Suresh        | Kalasalingam University, Krishnankoil. | Investigation of Various Multilevel Inverter topologies for Reactive Power Compensation of Distribution Power System”             | Completed |
| 6.  | D. Danalakshmi   | Kalasalingam University, Krishnankoil. | -“Reactive power Optimization and pricing in Regulated and Deregulated power System using Differential Evolution”                 | Completed |
| 7.  | K.Rajesh         | Kalasalingam University, Krishnankoil. | “Expansion Planning with Solar Power”                                                                                             | Completed |
| 8.  | R. Krishna Priya | Kalasalingam University, Krishnankoil. | “Integrated evolutionary fuzzy segmentation approach for detecting tumor in MR brain images”                                      | Completed |
| 9.  | B. Rampriya      | Kalasalingam University, Krishnankoil. | “Investigation of bidding strategies and profit maximization in unit commitment problem using differential evolution algorithms”, | Completed |
| 10. | M. Rajkumar      | Kalasalingam University, Krishnankoil. | “Combined economic emission dispatch using modified multi-objective genetic algorithm”                                            | Completed |
| 11. | S. Saravanan     | Kalasalingam University, Krishnankoil. | “Long term electricity consumption forecasting using computational intelligence techniques”                                       | Completed |
| 12. | T. Arun Prasath, | Kalasalingam University, Krishnankoil. | “Multi modal PET image reconstruction using soft computing techniques”                                                            | Completed |
| 13. | K. Karthikeyan   | Kalasalingam University, Krishnankoil. | “Application of Differential Evolution and its variants to Generation Expansion Planning”                                         | Completed |

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|-----|------------------------|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-------------|
| 14. | K. Vijayakumar,        | Kalasalingam University, Krishnankoil. | “Investigation of Multi-Function Grid Connected Converters for Enhanced Operation Employing Evolutionary Control Strategy” | Completed   |
| 15. | A. Arunkumar           | Anna University, Chennai.              | Eco-Friendly Generation Expansion Planning Considering Recuperation of Older Power Plants using PSO and its Variants       | Completed   |
| 16. | Sivapriya G            | Anna University                        | Design of Solar powered EV charging station using Machine learning based controller                                        | In Progress |
| 17. | Ms. Nazrin<br>Salma. S | Anna University                        | ABC                                                                                                                        |             |

#### 7.Short term Courses / Seminars /FDP/Workshop/ConferenceAttended:

| Sl.No | Title of the event                                                                                           | Organized by                                                                                             | Date       |            |
|-------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|------------|------------|
|       |                                                                                                              |                                                                                                          | From       | To         |
| 1.    | FDP on Clean Energy Technologies and Innovations in India: A Vision for Viksit Bhart @2047                   | IIITDM Kurnool, AICTE Training And Learning (ATAL) Academy                                               | 22.12.2025 | 27.12.2025 |
| 2.    | Attendance Certificate for “Academic Audit Committee member for Faculty Performance Appraisal process        | PACR Ramasamy Raja Polytechnic College, Rajapalayam                                                      | 11.10.2025 | -          |
| 3.    | “National Level Awareness Program on Institutional Development Plan (IDP) for Higher Education Institutions” | “Telangana Council of Higher Education (TGCHE), and Institute for Academic Excellence (IAE), Hyderabad.” | 06.03.2025 | -          |



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| 4.  | FDP - Advanced Technologies for Smart Power Systems                                                                                                               | "AICTE Training and Learning (ATAL) Academy at Noorul Islam Centre for Higher Education" | 10.02.2025 | 15.02.2025 |
| 5.  | “Extramural Lecture Series (ELS)- Mapping the CDIO Syllabus for Sustainability”                                                                                   | RIT - IQAC Ramco Institute of Technology                                                 | 20.12.2024 |            |
| 6.  | ATAL Basic FDP Recent Developments in IOT                                                                                                                         | Sankar PolyTechnic College Tirunelveli                                                   | 13.12.2024 |            |
| 7.  | Workshop - Impact of Research in Ranking and Accreditation                                                                                                        | Ramco Institute of Technology                                                            | 09.11.2024 |            |
| 8.  | Practical Hands on Ethical Hacking & Cyber Security Workshop (level 1)                                                                                            | Supraja Technologies                                                                     | 01.04.2024 | 06.04.2024 |
| 9.  | International Conference on Innovations in Computational intelligence and Computer vision "Customized Tricycle Assistive Technology for People With Disabilities" | RVS Technology Campus Coimbatore                                                         | 05.04.2024 | -          |
| 10. | International Conference on Innovations in Computational intelligence and Computer vision "Underground communication system for Mining safety"                    | RVS Technology Campus Coimbatore                                                         | 05.04.2024 | -          |
| 11. | "International Conference on ""Emerging trends in Science, Technology, Engineering and                                                                            | Bharath Niketan Engineering College Theni                                                | 26.04.2024 | 27.04.2024 |

Updated as on 02.04.2026

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|     | Management "" Smart shopping for visually impaired individuals”                                                                               |                                                                          |            |            |
| 12. | ""Extramural Lecture Series(ELS)- ELS002 - Integrating the CDIO Principles Through Effective Teaching-Learning Activities”"                   | Ramco Institute of Technology                                            | 29.02.2024 | -          |
| 13. | ""5th National Level Workshop on NIRF India Rankings 2024”"                                                                                   | Institute for Academic Excellence (IAE), Hyderabad.                      | 13.12.2023 | 14.12.2023 |
| 14. | ""Extramural Lecture Series (ELS)-ELS001 - Integrated Curriculum Development Methodology for Engineering using CDIO Framework: A Case Study”" | Ramco Institute of Technology                                            | 16.11.2023 | -          |
| 15. | "Train the Trainer program on Digital Productivity and AI Fluency under the Faculty Development Programme"                                    | Naan Mudhalvan in partnership with Microsoft                             | 06.10.2023 | 14.10.2023 |
| 16. | “ FDP -Design of Electrical Vehicles using MATLAB Simulink”                                                                                   | Ramco Institute of Technology<br>RIT Electra & IE (I) EE Student Chapter | 24.07.2023 | 26.07.2023 |
| 17. | “ FDP - Role of Artificial Intelligence in Power Sector”                                                                                      | PSN College of Engineering and Technology                                | 29.05.2023 | 02.06.2023 |
| 18. | Naan Mudhalvan Robotic Process Automation Foundation Course                                                                                   | Ui path                                                                  | 26.05.2023 | -          |
| 19. | International Conference on Recent Innovation in computer and manufacturing "IOT Based Food Waste manufacturing "                             | AAA College of Engineering Technology                                    | 12.04.2023 | 13.04.2023 |

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| 20. | “Research Proposal Writing”                                                                                                                        | Ramco Institute of Technology<br>- Research council,                                                            | 30.01.2023 | -          |
| 21. | National Assessment and Accreditation Council (NAAC)                                                                                               | Two day National level (online) Workshop on Role of NAAC in Enhancing Quality in Higher Education Institutions” | 16.12.2022 | 17.12.2022 |
| 22. | “ Workshop - TinyML for scientific use cases”                                                                                                      | Ramco Institute of Technology                                                                                   | 25.07.2022 | -          |
| 23. | International Conference on Smart Technologies and Applications-2022 (ICSTA 2022)                                                                  | Ramco Institute of Technology                                                                                   | 11.03.2022 | 12.03.2022 |
| 24. | STTP on Fundamentals of Electrical and Electronics (Tessolve Semiconductor Training – Phase I “Circuit Analysis                                    | RIT IE(I) EE                                                                                                    | 12.02.2022 | -          |
| 25. | "Renewable Energy – A future alternative"                                                                                                          | Government Polytechnic Thane AICTE Training and Learning (ATAL) Academy                                         | 17.01.2022 | 21.01.2022 |
| 26. | Outcome Based Learning"                                                                                                                            | Viveknand Education Society Business School AICTE Training and Learning (ATAL) Academy                          | 11.01.2022 | 15.01.2022 |
| 27. | International Conference ICPIC on Presented a paper on “ Prediction of Tamil Nadu’s Annual Electricity Demand Using Adoptive Neuro –Fuzzy Network” | PSN College of Engineering & Technology, Tirunelveli                                                            | 12.05.2021 | -          |
| 28. | Webinar on “Implementation of National Educational Policy – 2020 ”                                                                                 | RIT- IIC in association with RIT-IQAC & ISIE- RIT-SRA                                                           | 24.03.2021 | -          |
| 29. | Faculty Development Program on “NITTT - Orientation Training Programme for Mentors”                                                                | NITTTTR Chandigarh                                                                                              | 01.02.2021 | 05.02.2021 |

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| 30. | Online FDP on Electric Vehicles                                                                                                    | Maulana Azad National Institute of Technology, India                                   | 28.12.2020 | 02.01.2021 |
| 31. | (ATAL) Academy Online FDP on Waste Technology                                                                                      | C V Raman Global University                                                            | 21.12.2020 | 25.12.2020 |
| 32. | Short Term Training Program (STTP) on Embedding Soft Skills In Technical Education - Its Impact And Effectiveness on Employability | GRG Polytechnic College, Coimbatore                                                    | 17.11.2020 | -          |
| 33. | Orientation program on Patent Search and Drafting                                                                                  | RIT-ISTE -IIC                                                                          | 24.10.2020 | -          |
| 34. | FDP on Role of Faculty in Challenging Situation                                                                                    | RIT-ISTE                                                                               | 26.07.2020 | -          |
| 35. | Suddru sulalum manitha kula valvum                                                                                                 | Target Zero                                                                            | 05.06.2020 | 11.06.2020 |
| 36. | Interdisciplinary Webinar Series -I "Energy Audit and Management in Industrial Motors"                                             | Ramco Institute of Technology, Rajapalayam                                             | 02.06.2020 | -          |
| 37. | How to Write a Good Research Project proposal                                                                                      | Indian Institute of Technology Kanpur                                                  | 31.05.2020 | -          |
| 38. | What the Best College teachers do"                                                                                                 | RIT- Internal Quality Assurance cell -ISTE- Ramco Institute of Technology, Rajapalayam | 29.05.2020 | -          |
| 39. | Awareness on PANDEMIC Covid - 19"                                                                                                  | RIT-Faculty Recreation Club -ISTE- Ramco Institute of Technology, Rajapalayam          | 12.05.2020 | -          |
| 40. | Inculcating Universal Human Values in Technical Education                                                                          | All India Council for Technical Education                                              | 11.05.2020 | 15.05.2020 |
| 41. | Effective Use of Learning Management System                                                                                        | Ramco Institute of Technology, Rajapalayam                                             | 09.05.2020 | -          |
| 42. | Multi-Criteria decision- making workshop                                                                                           | REST Society for research International (RSRI                                          | 23.04.2020 | 25.04.2020 |

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| 43. | Artificial Intelligence                                                                                            | Indian Institute of Information Technology, Nagpur             | 24.04.2020 | 28.04.2020 |
| 44. | Leading in a VUCA World                                                                                            | ICT Academy - Sky Campus                                       | 20.04.2020 | 24.04.2020 |
| 45. | Yoga for Holistic Life Course                                                                                      | Temple of Consciousness, Aliyar.                               | 12.06.2019 | 16.06.2019 |
| 46. | Digsilent Power Factory                                                                                            | Ramco Institute of Technology, Rajapalayam.                    | 20.06.2016 | 22.06.2016 |
| 47. | FDP on Control System                                                                                              | Ramco Institute of Technology, Rajapalayam.                    | 04.06.2016 | 10.06.2016 |
| 48. | Art of Counselling                                                                                                 | Ramco Institute of Technology, Rajapalayam.                    | 19.06.2018 | 21.06.2018 |
| 49. | Energy Conservation through Water conservation”                                                                    | TANGEDCO (formerly TNEB)at Rajapalayam.                        | 18.12.2014 | -          |
| 50. | Energy conservation                                                                                                | TANGEDCO (formerly TNEB)at Rajapalayam.                        | 19.12.2013 | -          |
| 51. | Career Guidance Programme                                                                                          | Tamil Nadu Construction Workers Welfare Board at Virudhunagar. | 12.07.2012 | -          |
| 52. | Workshop on Solar Photovoltaics: Fundamentals, Technologies and Applications                                       | NIT, Tiruchirappalli.                                          | 12.12.2011 | 22.12.2011 |
| 53. | Mission10X Certification Workshop                                                                                  | Kalasalingam University, Krishnankoil.                         | 23.05.2011 | 24.05.2011 |
| 54. | Mission10X Faculty Empowerment Workshop                                                                            | Kalasalingam University, Krishnankoil.                         | 22.03.2011 | 26.03.2011 |
| 55. | TREC-STEP,                                                                                                         | NIT, Tiruchirappalli                                           | 10.03.2011 | -          |
| 56. | Faculty Advisor System, an orientation program organized for the new teachers by Centre for learning technologies, | CLT, Kalasalingam University Krishnankoil.                     | 08.07.2010 | -          |

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| 57. | Art of Counseling                                                                                             | Kalasalingam University, Krishnankoil.                                           | 27.04.2010 | 28.04.2010 |
| 58. | Conducted Academic audit conducted upon Faculty members                                                       | PSNA College of Engineering & Technology, Dindigul.                              | 09.09.2009 | -          |
| 59. | Concepts of Lithium Ion Battery                                                                               | Kalasalingam University, Krishnankoil.                                           | 03.09.2009 | -          |
| 60. | Statistical Modeling and Simulation                                                                           | Kalasalingam University, Krishnankoil.                                           | 01.06.2009 | 03.06.2009 |
| 61. | Mock Accreditation for Mechanical Engineering Department                                                      | Anand Institute of Higher Technology, Chennai.                                   | 07.04.2009 | 11.04.2009 |
| 62. | Engineering education in India? What it is & what it should be for the better employability of engineers      | Hotel Sangam, Madurai.                                                           | 23.03.2009 | -          |
| 63. | Insight 09                                                                                                    | Infosys Technologies Limited, Chennai                                            | 17.03.2009 | 18.03.2009 |
| 64. | 11 <sup>th</sup> ISTE Annual convention of Tamilnadu & Pondicherry Section & State level seminar.             | Mepco Schlenk Engineering College, Sivakasi                                      | 27.09.2008 | -          |
| 65. | ISTE Conference on “Industrial expectations and Institutional preparations”                                   | Sudharsan Engineering College, Sathiyamangalam, Pudukkottai District.            | 13.08.2008 | 14.08.2008 |
| 66. | LaTeX (document preparation system)                                                                           | n-CARDMATH and ISTE staff chapter of Kalasalingam University, Krishnan Koil.     | 23.07.2008 | -          |
| 67. | Weekly Seminars                                                                                               | Power and Energy system group, Department of Electrical and Computer Engineering | 10.10.2006 | 29.09.2007 |
| 68. | Pre-conference tutorial on, National Power Engineering Conference- NPEC- 2005-Research topics in Power System | Department of EEE, Thiagarajar College of Engineering, Madurai.                  | 04.12.2005 | 15.12.2005 |

Updated as on 02.04.2026

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|     | Restructuring Environment                                                                          |                                                                                 |            |            |
| 69. | S.P.I.D.E.R. RTU 232 & MicroSCADA                                                                  | ABB Training Centre, Bangalore.                                                 | 15.04.2003 | 17.04.2003 |
| 70. | Training program on ETAP, PSCAD/EMTDC software                                                     | Kalasalingam University, Krishnankoil.                                          | 11.12.2002 | 13.12.2002 |
| 71. | Short term training Programme on “Genetic Algorithms in Engineering Applicatins                    | Coimbatore Institute of Technology, Coimbatore.                                 | 19.11.2001 | 02.12.2001 |
| 72. | Faculty development Programme for Computer simulation of Electrical Power System                   | Mepco Schlenk Engineering College, Sivakasi.                                    | 21.05.2001 | 22.05.2001 |
| 73. | Fuzzy Logic & Neural Network                                                                       | Mohamed Sathak Engineering College, Kilakarai.                                  | 23.02.2001 | 24.02.2001 |
| 74. | AICTE-ISTE Winter School on Reliability Engineering and Applications to Power Systems (REAPS-2000) | J.N.T.U. College of Engineering, (Autonomous College), Anantapur, Andrapradesh. | 11.12.2000 | 23.12.2000 |
| 75. | Two day Workshop on Neuro-Fuzzy-Genetic Systems                                                    | Kumaraguru College of Technology, Coimbatore –                                  | 03.03.2000 | 04.03.2000 |
| 76. | Faculty development Programme in Innovation and Entrepreneurship                                   | Regional Engineering College, Tiruchirappalli                                   | 18.01.2000 | 02.02.2000 |

#### 8. Short term Courses / Seminars /FDP/Workshop/Conference organized:

| Sl.No | Title of the event                                                                                                             | Organized by | Date       |            |
|-------|--------------------------------------------------------------------------------------------------------------------------------|--------------|------------|------------|
|       |                                                                                                                                |              | From       | To         |
| 1.    | Six-Day Short-Term Training Programme on "Emerging Trends in Smart E-Mobility and Sustainable Electric Powertrain Technologies | ISTE- RIT    | 15.12.2025 | 20.12.2025 |
| 2.    | FDP on AI and ML for electrical engineering                                                                                    | RIT          | 05.05.2025 | 09.05.2025 |

Updated as on 02.04.2026

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| 3.  | Guest Lecture on "Battery Management System for Electric Vehicle Applications"                     | RIT                  | 22.03.2025                             | -          |
| 4.  | FDP on Competency based Engineering Education                                                      | EEE & MECH - RIT     | 10.02.2025                             | 15.02.2025 |
| 5.  | Work shop on "Modern Electrical Measurement & Sustainable Power Analysis for Electrical Engineers" | IE(I) – RIT          | 05.02.2025                             | -          |
| 6.  | A 6-day Value added Course on "Selection and Testing of Industrial Sensors"                        | RIT                  | 20.01.2025                             | 25.01.2025 |
| 7.  | A 3-day Value added Course on "Creative Electronics with Arduino for Practical Applications"       | RIT                  | 20.01.2025                             | 22.01.2025 |
| 8.  | Workshop on "MATLAB for Electrical Engineering Applications"                                       | IE(I) – RIT          | 21.02.2024                             | -          |
| 9.  | Short Term Training Programme on Fundamentals of Electrical and Electronics                        | IE(I) – RIT          | 10.06.2024<br>15.06.2024<br>16.06.2024 |            |
| 10. | Workshop on 'Mastering Technical Writing: From Basics to Ethical AI and illustrations'             | IE(I) – RIT          | 19.02.2024                             | -          |
| 11. | Ethical Hacking & Cyber Security Workshop (Level -1)                                               | Supraja Technologies | 01.04.24                               | 06.04.24   |
| 12. | Play with PIC                                                                                      | RIT                  | 14.11.23                               | -          |
| 13. | Overview of Semiconductor Test Industry                                                            | RIT                  | 09.11.23                               | -          |
| 14. | Understanding the Datasheet in Test Engineer Prospective                                           | RIT                  | 09.11.23                               | -          |
| 15. | Opportunities for higher education in India                                                        | RIT                  | 29.09.23                               | -          |



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| 16. | How to handle interviews in the software sector                                                                       | RIT                                                       | 29.09.23           | -          |
| 17. | Semiconductor Test Engineering                                                                                        | RIT                                                       | 29.09.23           | -          |
| 18. | How to face the interviews in the software industry                                                                   | RIT                                                       | 27.09.23           | -          |
| 19. | Online Guest Lecture on the topic “Model, simulate and analyze the performance characteristics of Lead Acid Batteries | IE(I) EE Student chapter                                  | 22.09.2023         | -          |
| 20. | Three Day Faculty Development Program on “Design of Electric Vehicle using MATLAB Simulink”                           | RITELECTRA technical association IE(I) EE Student Chapter | 24.07.2023         | 26.07.2023 |
| 21. | International Conference on Smart Engineering for Renewable Energy Technologies (ICSERET 2023)                        | RIT                                                       | 24.03.2023         | 25.03.2023 |
| 22. | UPSC and TNPSC Civil Services Examinations                                                                            | RIT<br>CGC                                                | 07.02.2023         | -          |
| 23. | Srinivasa Ramanujan Mathematical Competition for School Students – 2022                                               | ISTE                                                      | 27.01.2023         | -          |
| 24. | Higher Education in Abroad                                                                                            | RIT<br>CGC                                                | 01.10.2022         | -          |
| 25. | Wind Power – Growth and Opportunities in India                                                                        | RIT<br>CGC                                                | 24.09.2022         | -          |
| 26. | Vayumitra STTP - O&M Electrical & Instrumentation Technician – Wind Power Plant”                                      | NIWE, MHRD, RIT                                           | 80 hours per batch | -          |
| 27. | Workshop on “TinyML for scientific use cases”                                                                         | RIT                                                       | 25.07.2022         | -          |
| 28. | Book review session                                                                                                   | RIT                                                       | Every Sunday       | -          |

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| 29. | FDP on Bio Inspired Optimization Techniques for Electrical Engineering Application on “ Genetic Algorithm (GA)with hands-on training” | RIT – Resource Person   | 21.02.2022            | 23.02.2022 |
| 30. | International Conference on Smart Technologies and Applications                                                                       | AICTE, NABARD, DBT, BAI | 11.03.2022            | 12.03.2022 |
| 31. | Hands on Training on "Project using Wireless devices                                                                                  | RIT                     | 02.11.2021            | -          |
| 32. | Webinar on placement opportunities for EEE students in "Core sector                                                                   | RIT                     | 24.09.2021            | -          |
| 33. | Short Term Training Programme on Build a Team                                                                                         | RIT                     | 23.10.2021            | -          |
| 34. | Webinar on placement opportunities for EEE students in "Soft sector"                                                                  | RIT                     | 22.09.2021            | -          |
| 35. | Webinar on opportunities in "Higher Education" for EEE students                                                                       | RIT                     | 21.09.2021            | -          |
| 36. | "Opportunities for Higher Education in Abroad"                                                                                        | RIT CGC                 | 20.09.2021            | -          |
| 37. | "Huge Opportunities and Emerging Newer Scope for All Core Branch Talented Engineers in India                                          | RIT CGC                 | 29.06.2021            | -          |
| 38. | Webinar on "How to crack TCS Ninja"                                                                                                   | RIT                     | 08.08.2021<br>3 Hours | -          |
| 39. | Seminar on Stress Management during COVID                                                                                             | RIT & Arvind Herbals    | 16.07.2021            | -          |
| 40. | Online Hands on Training on the topic “Office Automation                                                                              | RIT                     | 04.06.2021            | 05.06.2021 |
| 41. | Online Course on "Sensors and Displays"                                                                                               | RIT                     | 26.05.2021            | 27.05.2021 |

Updated as on 02.04.2026

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| 42. | Online Course on the topic "Fun with Electric Circuits"                             | RIT                                                                                                                  | 12.05.2021<br>19.05.2021 | 13.05.2021<br>20.05.2021 |
| 43. | National Conference 2021                                                            | RIT                                                                                                                  | 19.03.2021               | 20.03.2021               |
| 44. | Webinar on Empower Employment training programme on "How to Crack Aptitude"         | Alumni Association                                                                                                   | 06.01.2021               | -                        |
| 45. | Webinar on the topic "Motivational Session on Placement Recruitment Drive"          | IEEE Student Branchlike(I) Student Chapter                                                                           | 02.01.2021               | -                        |
| 46. | Group Discussion                                                                    | RIT                                                                                                                  | 31.12.2020<br>Half Day   | -                        |
| 47. | Webinar on Empower Employment training programme on "How to Crack Aptitude"         | Association with IEEE Student Branch, IE(I) Student Chapter                                                          | 29.12.2020               | -                        |
| 48. | Webinar on Empower Employment training programme on "How to Crack Aptitude"         | Association with IEEE Student Branch, IE(I) Student Chapter                                                          | 22.12.2020               | -                        |
| 49. | value added course on "EVA039 Virtual Instrumentation"                              | RIT                                                                                                                  | 29.11.2020               | 05.12.2020               |
| 50. | Webinar on "Empower your skills on Electrical Machines"                             | ISIE-RITSAR                                                                                                          | 19.11.2020.              | -                        |
| 51. | Workshop on "Virtual Laboratory                                                     | RIT- ELECTRA, IEEE Student Branch, IE(I) EE                                                                          | 29.10.2020               | -                        |
| 52. | Webinar on "Innovative Ideas for Projects and Guidelines for MSME project proposal" | RIT-ELECTRA, IEEE Student Branch, IE(I) EE Student Chapter                                                           | 11.10.2020               | -                        |
| 53. | Industry Lecture on "Energy Conservation Activities in Industry Premises            | Energy Conservation Quality Circle (ECQC) in Association with RIT- EEE Alumni Association and RITIEEE Student Branch | 04.10.2020<br>2 Hours    | -                        |

Updated as on 02.04.2026

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|-----|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------|------------|
| 54. | Webinar on the topic "Digital Electronics"                                                   | RIT Training placement Centre, RITEEE Alumni Association, RIT - IE(I) | 20.09.2020            | -          |
| 55. | Technical Quiz on "Fundamentals of Electronics"                                              | RIT                                                                   | 15.09.2020<br>2 day   | -          |
| 56. | Design Thinking and Innovation                                                               | RIT                                                                   | 29.08.2020<br>2 Hours | -          |
| 57. | Webinar on How to face an Interview and Technical requirement for different IT job portfolio | RIT (IE)                                                              | 23.08.2020            | -          |
| 58. | IWCSET 2020                                                                                  | RIT                                                                   | 29.06.2020            | 30.06.2020 |
| 59. | Interdisciplinary Webinar Series-II                                                          | RIT                                                                   | 08.06.2020            | 12.06.2020 |
| 60. | Interdisciplinary Webinar Series – I                                                         | RIT                                                                   | 01.06.2020            | 05.06.2020 |
| 61. | Research Article and Reference Management using Zotero                                       | RIT- RC                                                               | 29.05.2020            | -          |
| 62. | Sensors and Displays                                                                         | RIT                                                                   | 25.05.2020            | 26.05.2020 |
| 63. | How to formulate a Research Problem                                                          | RIT- RC                                                               | 13.05.2020            | -          |
| 64. | Fun with Electric Circuits                                                                   | RIT                                                                   | 11.05.2020            | 13.05.2020 |
| 65. | Effective Use of Learning Management System                                                  | RIT- RC                                                               | 09.05.2020            | -          |
| 66. | Introduction to IPR                                                                          | RIT- RC                                                               | 23.03.2020            | -          |

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| 67. | One day workshop on Arduino (during TECHYUGA)                                                                        | RIT                                                              | 31.08.2019 | -          |
| 68. | Energy Conservation Awareness Program                                                                                | RIT-ECQC                                                         | 29.08.2019 | -          |
| 69. | Writing Strategy of Research Proposal                                                                                | RIT- RC                                                          | 12.01.2019 | -          |
| 70. | one day workshop titled “MATLAB Onramp”                                                                              | Ramco Institute of Technology                                    | 22.11.2018 | -          |
| 71. | Faculty Development Training Programme on “Signals and Systems”, organized by Dept. of ECE & EEE                     | Ramco Institute of Technology.                                   | 13.06.2018 | 18.06.2018 |
| 72. | AICTE sponsored Two-week Faculty Development Programme (FDP) on “Human Values and Professional Ethics”               | Ramco Institute of Technology.                                   | 30.10.2017 | 12.10.2017 |
| 73. | Faculty Development Training Programme on IC 6501 - Control Systems                                                  | Centre for Faculty Development, Anna University, Chennai.        | 04.06.2016 | 10.06.2016 |
| 74. | One day National Level workshop on “Emerging Trends in Industrial Safety Engineering (ETISE-11)”                     | Kalasalingam University, Anand Nagar, Krishnankoil.              | 14.10.2011 | -          |
| 75. | National Conference on Innovations in Engineering, Science and Technology - NCIEST 2015                              | Ramco Institute of Technology, Rajapalayam.                      | 06.03.2015 | 07.03.2015 |
| 76. | International Conference on Artificial Intelligence and Evolutionary Algorithms in Engineering Systems (ICAEES-2014) | Centre for Higher Education, Kumaracoil, Thuckalay, Kanyakumari. | 22.04 2014 | -          |
| 77. | International Conference on Power and Energy System                                                                  | Velammal College of Engg.and Technology, Madurai.                | 20.12.2013 | -          |
| 78. | International Conference on Power & Energy Systems                                                                   | ICPES 2012, KLU                                                  | 13.12.2012 | 15.12.2012 |

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| 79. | National conference on “New and Renewable Energy” – NRE’12                        | Einstein College of Engineering, Tirunelveli.                   | 25.02.2012 | -          |
| 80. | National conference on Intelligent Computing in Communication and Automation”     | Kalasalingam University, Krishnankoil.                          | 08.04.2011 |            |
| 81. | “National Conference on Power and Energy Systems” (NPES’11)                       | Kalasalingam University, Krishnankoil                           | 18.03.2011 | 19.03.2011 |
| 82. | Technorions 1.0                                                                   | Kalasalingam Institute of Technology, Anand Nagar, Krishnankoil | 24.09.2010 | -          |
| 83. | “National Conference on Power and Energy Systems” (NPES’10)                       | Kalasalingam University, Krishnankoil.                          | 19.03.2010 | 20.03.2010 |
| 84. | National Conference on Recent Trends in Communication Systems – 2010 (NCRCS-2010) | Kalasalingam University, Krishnankoil.                          | 11.03.2010 | 12.03.2010 |
| 85. | National level Technical symposium, ENITUXIA 2K10                                 | Theni Kammavar Sangam College of Technology, Theni.             | 26.02.2010 | -          |
| 86. | International Conference on Advanced Manufacturing and Automation(INCAMA 2009)    | Kalasalingam University, Krishnankoil.                          | 28.03.2009 | -          |
| 87. | National level Technical Symposium, EUPHEMOZ-08                                   | Kalasalingam University, Krishnankoil.                          | 19.09.2008 | -          |
| 88. | State level Technical Symposia                                                    | Arulmigu Kalasalingam Polytechnic College, Krishnankoil         | 30.08.2008 | -          |
| 89. | National conference on “Power Electronics & Power Systems (MEPPCON’08)            | Mepco Schlenk Engineering College, Sivakasi                     | 14.03.2008 | -          |
| 90. | National level Technical symposium, ELESYTEC’06                                   | Thiagarajar College of Engineering, Madurai                     | 20.09.2006 | -          |
| 91. | National level students Technical symposia                                        | P.S.R. Engineering College, Sevalpatti                          | 07.09.2006 | -          |

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| 92. | National conference on “Exploring New Frontier in Power Conversion Technologies(PCTCON’06)    | P.S.N.A. College of Engg. and Tech., Dindigul | 26.04.2006 | -          |
| 93. | National Conference on “Intelligent Computing in Communication and Automation”, (NCICCA-2005) | Kalasalingam University, Krishnankoil.        | 01.04.2005 | 02.04.2005 |

## 9. online Courses

| sno | Course title                                              | University                                              | Date       |
|-----|-----------------------------------------------------------|---------------------------------------------------------|------------|
| 1.  | The Science of Well-Being                                 | Yale University and offered through Coursera            | 24.02.2025 |
| 2.  | Data Science Math Skills                                  | Duke University and offered through Coursera            | 20.11.2024 |
| 3.  | "Mathematics for Machine Learning: Multivariate Calculus" | "Imperial College London and offered through Coursera"  | 05.04.2024 |
| 4.  | Battery comparison manufacturing and Packaging            | Arizona State University<br>Offered through by Coursera | 10.01.2024 |
| 5.  | Differential Equation for Engineers                       | The Hong Kong University Science and technology         | 24.06.2023 |
| 6.  | Getting Started with Deep Learning                        | DEEP LEARNING INSTITUTE                                 | 25.11.2022 |
| 7.  | "Computational Thinking Training"                         | TCS EMPOWERS                                            | 10.10.2022 |
| 8.  | Mathematics for Machine Learning: Linear Algebra          | " Imperial College London and offered through Coursera" | 25.02.2022 |
| 9.  | MATLAB Onramp                                             | Maths works Training Services                           | 25.01.2022 |
| 10. | Mastering Programming with MATLAB                         | Vanderbilt University                                   | 16.09.2021 |
| 11. | Linear Circuits 1: DC Analysis                            | Georgia Institute of Technology                         | 21.09.2020 |
| 12. | Operations Research: an Active Learning Approach          | The Hong Kong Polytechnic University                    | 03.09.2020 |
| 13. | Transfer functions and Laplace transforms                 | MIT                                                     | 01.09.2020 |

Updated as on 02.04.2026

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|-----|---------------------------------------------------------------------------|----------------------------------------------------|------------|
| 14. | Programming for Everybody (Getting Started with Python)                   | University of Michigan                             | 31.07.2020 |
| 15. | Electric Cars: Introduction                                               | Delft University of Technology                     | 23.07.2020 |
| 16. | Electric Power Systems                                                    | The State University of Newyork                    | 13.07.2020 |
| 17. | Vector Calculus for Engineers                                             | The Hong Kong University of Science and Technology | 17.06.2020 |
| 18. | Introduction to Differential equations                                    | MIT                                                | 26.12.2019 |
| 19. | Circuits and Electronics 2: Amplification, Speed, and Delay               | MITx                                               | 31.05.2019 |
| 20. | Calculus 1C: Coordinate Systems & Infinite Series                         | MITx                                               | 12.05.2019 |
| 21. | Differential Equations: Fourier Series and Partial Differential Equations | MITx                                               | 21.12.2018 |
| 22. | Calculus 1B: Integration                                                  | MITx                                               | 09.03.2018 |
| 23. | Calculus 1A: Differentiation                                              | MITx                                               | 10.09.2017 |
| 24. | Electricity and Magnetism, Part 1                                         | Rice University                                    | 01.06.2017 |
| 25. | Introduction to Programming with MATLAB                                   | Vanderbilt University                              | 24.09.2016 |
| 26. | Circuits and Electronics 1: Basic Circuit Analysis                        | MITx                                               | 09.05.2016 |

#### 10. Patents filed/awarded:

| S. NO. | NAME OF THE MEMBERS                                                                         | NAME OF THE PATENT                                           | PATENT APPLIED NUMBER & DATE     | CURRENT STATUS |
|--------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------|----------------------------------|----------------|
| 1.     | S, Periyamayagi<br>S.Kannan<br>B.Deepa Lakshmi,<br>K., Karthikeyan and<br>Moorthi Kanagaraj | Smart Uniform for Girl Children with Artificial Intelligence | 202010289620<br>&<br>October2020 | Granted        |

Updated as on 02.04.2026



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|----|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-------------------------------------|
| 2. | Ms.M.Kirubah<br>Ms.K.Sivapriya<br>Ms.G.Yashaswini<br>Mr. E. Thangam<br>Mr. N. Ganesh<br>Dr. K. Karthikeyan<br>Dr.S.Kannan                    | Method for<br>Treating<br>Waste Water<br>in Dye<br>Industry using<br>Natural Filter                                                                 | 202041056624<br>&<br>28.12.2020 | FER received<br>and<br>communicated |
| 3. | Mrs. S. Jeyanthi<br>Mr. A. Arun Kumar<br>Dr. S. Kannan<br>Dr. G. Kanthimathi<br>Dr. O. Senthil Kumar                                         | MultiTerminal<br>CorrosioFree<br>Service Line<br>Clamp                                                                                              | 202141004396<br>&<br>01.02.2021 | Granted                             |
| 4. | Ms. S. Jeya Shiny<br>Suhel<br>Ms. M. Ramya<br>Ms. A. G. Vinataa<br>Ms.S.Sharmila<br>Kumari<br>Mrs G.Sivapriya<br>Mr.E.Thangam<br>Dr.S.Kannan | Apparatus for<br>Monitoring<br>and<br>Dispensing<br>Medicine to<br>Elderly<br>Patients                                                              | 202141004468<br>&<br>02.02.2021 | FER received<br>and<br>communicated |
| 5. | Mr. D. Logesh<br>Mr. C. Kaleeshwaran<br>Mr. M. Arumugaananth<br>Dr.S.Kannan<br>Ms.S.Sharmila Kumari<br>Mrs G.Sivapriya                       | Improving the<br>Efficiency of<br>Off-Grid<br>Bifacial Solar<br>Panel using<br>Concentrator<br>or Reflector<br>System                               | 202241018277<br>&<br>28.03.2022 | Granted                             |
| 6. | P.Vishnuram<br>S.Manikandan<br>Dr.S.Kannan<br>Mr.A.Guna,<br>Mr.D.Karthik Prabhu,                                                             | State of<br>charge based<br>automatic<br>charging<br>controller with<br>temperature<br>monitoring<br>system for<br>Electric<br>Vehicle<br>batteries | 202241043872<br>&<br>01.08.2022 | FER received<br>and<br>communicated |
| 7. | Mr. K.K.Kishore<br>Mr. K. Mubis Akash<br>Varun<br>Mr. R.K. Venkateshan<br>Dr.S.Kannan<br>Mrs G.Sivapriya<br>Ms.S.Sharmila Kumari             | An Internet of<br>Things System<br>for Rainwater<br>Collection and<br>Intelligent<br>Farming                                                        | 202441015382<br>&<br>01.03.2024 | Published<br>08.03.2024             |

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|-----|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------|----------------------|
| 8.  | Dr. S .Kannan<br>Dr. D. Karthik Prabhu,<br>Mr. E. Thangam,                                                            | Autonomous Agriculture Robot for precision Farming                                | 202441037390<br>Docket No. 67000; CBR No. 31851 & 13.05.2024 | Published 17.05.2024 |
| 9.  | E. Thangam<br>P. Prem<br>D. Karthik Prabhu<br>S. Vijay Anandh<br>U. Ponselvam<br>S. K. Nitheesh<br>Arjun<br>S. Kannan | A System and Method for Testing Paper Currency                                    | 29.05.2024                                                   | Filed                |
| 10. | SIVAPRIYA G,<br>S.Sharmila Kumari,<br>Prem, Dr. S.<br>KANNAN                                                          | Detachable electricmotor drive forWheelchair                                      | 202441063082                                                 | Published 21-08-2024 |
| 11. | Dr. S. KANNAN                                                                                                         | Lake Guardian :Autonomous RoboticBoats For DebrisCleanup and Pollution Prevention | 202541057403                                                 | Published 16-06-2025 |

### 11. Membership in Professional societies

Life member in Indian Society for Technical Education (ISTE) (LM 28384), since 2000

ii) Institution of Engineers (India) Limited, Kolkata, India.

- ✓ Associate Member - (AM/088422-5), with effect from 22.01.2002 to 15.02.2009.
- ✓ Member - (M/138243/2 EL), with effect from 16.02.2009 to 19.09.2010.
- ✓ Fellow - (F-115344-3), with effect from 20.09.2010.
- ✓ Chartered Engineer (India), with effect from 26.06.2012.

Attached to Tamilnadu state centre and Madurai local centre

iii) Senior Member (No. 80760671) in Institution for Electrical and Electronics Engineers (IEEE), USA and Member, Power Engineering Society, IEEE, USA, since 2007. (M'2007, SM'2011)

iv) Senior Member in Computer Society of India – (Member # 00173867), since 1<sup>st</sup> Nov. 2008.

v) Fellow in Institution of Electronics and Telecommunication Engineers, New Delhi, India. (F-209728), with effect from 19<sup>th</sup> Jan. 2009.

vi) Life member (LM No. 30929), Systems Society of India (SSI), since Oct. 2010.

**Updated as on 02.04.2026**

vii) Member (No. 1100321947) in IET, The Institute of Engineering and Technology, since Jan. 2014

## **12. Awards, Recognition & Achievements**

- a. Special Appreciation for his Effort as Coordinator and Conducted SRMC (Srinivasa Ramanujam Mathematical Competitions for the School Students in Virudhunagar District for the Academic Year 2021-22.
- b. An Subject for 100 percent Result in December – January 2019,
- c. As, an advisor for RIT- IEEE Student branch, achieved an appreciation certificate award from IEEE Madras Section – Best Student branch for the year 2019
- d. National rural merit scholarship in district level (Ramanathapuram) in 8<sup>th</sup> standard for excellent performance in studies. Received a financial assistance of Rs. 1000 per year for four years from 9<sup>th</sup> standard to 12<sup>th</sup> standard.
- e. Better Opportunities for Young Scientists In Chosen Areas of Science and Technology (BOYSCAST) fellowship for the year 2005-2006, awarded by Department of Science and technology (DST), Govt. of India, New Delhi and carried out research during 10, Oct. 2006 to 29, Sep. 2007, in “Multicriteria decision making including Genetic Algorithms and Neural networks” under the supervision of Dr. James D. McCalley, Professor, Electrical and Computer Engineering, Iowa State University, Ames, Iowa, USA.
- f. Representative - from Research and Educational Institution – Demand Side Management Consultation Committee constituted by Tamilnadu Electricity Regulatory Commission for a period of 03 years (25.06.2014 onwards)
- g. ISTE Periyar award for Best Engineering College, for the year 2024, organized by Indian Society for Technical Education Tamilnadu section -R.M.K Engineering College